ME416
Ground Vehicle Dynamics

Performance - Acceleration

Fall 2018

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Introduction

- The performance of a vehicle is the common term used to refer to a vehicle's
  - Acceleration (Time To Speed)
  - Top speed
  - Gradability
  - Braking
- These performance values are dependent upon many factors
  - Vehicle mass
  - Weight distribution
  - c.g. height
  - Tire/surface grip characteristics
  - Engine torque and speed characteristics
  - Transmission ratios
  - FWD, RWD, 4WD (or xWD)
  - Aerodynamic characteristics
  - Rotational inertia of rotating parts
  - and so on…

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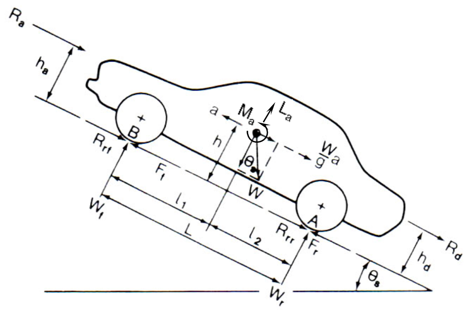
Acceleration in a Straight Line



- The ability of a vehicle to accelerate in a straight line is limited by
  - The available power from the engine
  - The traction available at the driven wheels
- The forces acting on the car can be represented by the FBD shown to the left

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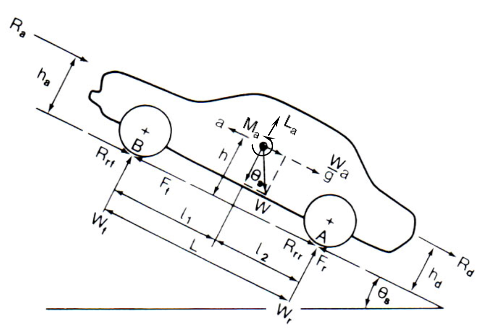
Dynamic Axle Loads



- Note that the weight of the car on level ground is simply the mass times gravity but on a grade will consist of a cosine term perpendicular to the road and a sine term parallel to the road
- The tires are shown supporting a normal load – this load can consist of
  - The static weight components as a function of the static weight distribution
  - Trailer tongue normal load (not shown)
  - Aerodynamic lift (downforce) and pitching moment
  - A dynamic load component, i.e. weight transfer, due to acceleration

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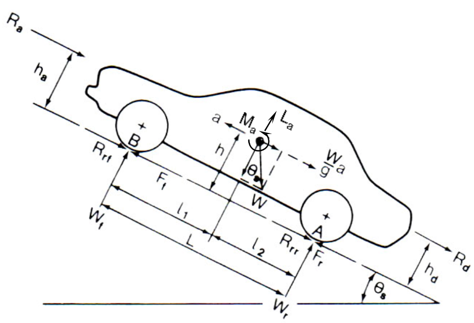
Longitudinal Load Transfer



- Assume
  - Grade angle is small
  - All tires are similar in construction, material and inflation
  - No drawbar normal load, i.e. tongue weight is zero
  - Rotational inertia of driveline ignored, i.e. transmission, driveshaft, etc.
  - Lateral load transfer due to applied torques ignored

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Longitudinal Load Transfer (2)

- We can sum the forces for the system

$$\sum F_x = ma = (F_{Tf} + F_{Tr}) - (R_{rr} + R_{rf}) - R_a - R_d - W\sin\theta_s$$

and the moment about the front axle

Accounts for aerodynamic lift and moment: $M_a = L_a(l_1 + l_{cp})$ where l_{cp} is the point of action of the lift force

$$(+)\sum M_f = hma + 4I_T\alpha = -h_aR_a - hW\sin\theta_s - l_1W\cos\theta_s + w_rL - h_dR_d + L_a(l_1 + l_{cp})$$

and the moment about the rear axle

Assumes aerodynamic drag, R_a, acts at height h_a

$$(+)\sum M_r = hma + 4I_T\alpha = -h_aR_a + hW\sin\theta_s + l_1W\cos\theta_s - W_fL - h_dR_d + L_a(l_2 - l_{cp})$$

Note the aerodynamic forces as treated by Iannce and Wong - this is most likely not the most convenient or standard format depending on how your aerodynamic data is presented

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Longitudinal Load Transfer (3)

- Assume
  - Grade angle is small
  - All tires are similar in construction, material and inflation
  - No drawbar normal load, i.e. tongue weight is zero
  - Rotational inertia of driveline ignored, i.e. transmission, driveshaft, etc.
  - Lateral load transfer due to applied torques ignored

$$\oplus\sum M_f = hma + \cancelto{0}{4I_f\alpha} = -W_r L - h_a R_a - hW \cancelto{\theta_s}{\sin\theta_s} - l_1 W \cancelto{1}{\cos\theta_s} - h_d R_d + l_1 L_a + M_a$$

Aerodynamic Forces at c.g

$$\oplus\sum M_r = hma + \cancelto{0}{4I_r\alpha} = -W_f L - h_a R_a - hW \cancelto{\theta_s}{\sin\theta_s} + l_1 W \cancelto{1}{\cos\theta_s} - h_d R_d + l_2 L_a + M_a$$

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Longitudinal Load Transfer (4)

- Subtracting yields

$$0 = (w_f + w_r)L - (l_1 + l_2)W + (l_1 + l_2)L_a$$

setting $(l_1+l_2)=L$

$$W_f + W_r = W - L_a$$

- The rolling resistance is given by

$$R_r = R_{rf} + R_{rr} = \mu_r W_f + \mu_r W_r = \mu_r(W_f + W_r)$$
$$= \mu_r(W - L_a)$$

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Longitudinal Load Transfer (5)

- Solving the force equation for the net tractive force

$$\sum F_x = ma = (F_{Tf} + F_{Tr}) - (R_{rr} + R_{rf}) - R_a - R_d - W\sin\theta_s$$

where

$$F_{T,Net} = F_T - R_r$$

yields

$$F_{T,Net} = -R_a - R_d - W\underset{\theta_s}{\cancel{\sin\theta_s}} - ma$$

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Longitudinal Load Transfer (6)

- Now, solving the moment equations for W_f and W_r

$$\sum M_f = hma + \underset{0}{\cancel{4I_T\alpha}} = -W_r L - h_a R_a - hW\underset{\theta_s}{\cancel{\sin\theta_s}} - l_1 W\underset{1}{\cancel{\cos\theta_s}} - h_d R_d + l_1 L_a + M_a$$

$$W_r = \frac{1}{L}\left[hma + h_a R_a + hW\theta_s + l_1 W + h_d R_d - l_1 L_a - M_a\right]$$

$$\sum M_r = hma + \underset{0}{\cancel{4I_T\alpha}} = -W_f L - h_a R_a - hW\underset{\theta_s}{\cancel{\sin\theta_s}} + l_2 W\underset{1}{\cancel{\cos\theta_s}} - h_d R_d + l_2 L_a + M_a$$

$$W_f = \frac{1}{L}\left[-hma - h_a R_a - hW\theta_s + l_2 W - h_d R_d - l_2 L_a + M_a\right]$$

- Assume the drawbar load acts through the c.g. (for simplicity)

$$W_f = \frac{1}{L}\left[l_2(W - L_a) + M_a - h(R_a + R_d + W\theta_s + ma)\right]$$

$$W_r = \frac{1}{L}\left[l_1(W - L_a) - M_a + h(R_a + R_d + W\theta_s + ma)\right]$$

$F_{T,Net} = F_T - R_r$

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Maximum Available Traction – 4WD

- Assuming there is no wheel slip the maximum available traction for a 4WD car is the total friction force available

$$
\begin{aligned}
F_{T,max,4WD} &= \mu_T (W_f + W_r) \\
&= \mu_T \left(\frac{l_2}{L} + \frac{l_1}{L} \right) (W - L_a) \\
&= \mu_T (W - L_a)
\end{aligned}
$$

  - This equation tells us nothing about the actual torque split front to rear for maximum performance due to dynamic load transfer
  - We can define a tractive efficiency

$$
\frac{F_{T,max,4WD}}{\mu_T W} = 1 - \frac{L_a}{W}
$$

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Maximum Available Traction - RWD

- Maximum available traction for a RWD car

$$
\begin{aligned}
F_{T,max,RWD} &= \mu_T W_r \\
&= \frac{\mu_T}{L} \left[l_1 (W - L_a) - M_a + h(R_a + R_d + W\theta_s + ma) \right] \\
&= \frac{\mu_T}{L} \left[l_1 (W - L_a) - M_a + h(F_{T,max} - R_r) \right] \\
&= \frac{\frac{\mu_T}{L} \left[(l_1 - h\mu_r)(W - L_a) - M_a \right]}{\left(1 - \mu_T \frac{h}{L} \right)}
\end{aligned}
$$

  - We can define a tractive efficiency

$$
\frac{F_{T,max\,RWD}}{\mu_T W} = \frac{\left(\frac{l_1}{L} - \frac{h\mu_r}{L} \right)\left(1 - \frac{L_a}{W} \right) - \frac{M_a}{LW}}{\left(1 - \mu_T \frac{h}{L} \right)}
$$

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Maximum Available Traction - FWD

- Maximum available traction for a FWD car

$$
\begin{aligned}
F_{T,\max,FWD} &= \mu_T W_f \\
&= \frac{\mu_T}{L}\left[l_2(W - L_a) + M_a - h(R_a + R_d + W\theta_s + ma)\right] \\
&= \frac{\mu_T}{L}\left[l_2(W - L_a) + M_a - h(F_{T,\max} - R_r)\right] \\
&= \frac{\frac{\mu_T}{L}\left[(l_2 + h\mu_r)(W - L_a) + M_a\right]}{\left(1 + \mu_T \frac{h}{L}\right)}
\end{aligned}
$$

  - We can define a tractive efficiency

$$
\frac{F_{T,\max FWD}}{\mu_T W} = \frac{\left(\frac{l_2}{L} + \frac{h\mu_r}{L}\right)\left(1 - \frac{L_a}{W}\right) + \frac{M_a}{LW}}{\left(1 + \mu_T \frac{h}{L}\right)}
$$

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Maximum Available Traction

- Maximum available tractive efficiency
  - 4WD

$$
\frac{F_{T,\max,4WD}}{\mu_T W} = 1 - \frac{L_a}{W}
$$

Assuming a correct F/R torque split then only aerodynamic lift affects the tractive efficiency. Downforce enhances the tractive efficiency for all drive configurations.

  - RWD

$$
\frac{F_{T,\max RWD}}{\mu_T W} = \frac{\left(\frac{l_1}{L} - \cancel{\frac{h\mu_r}{L}}\right)\cancel{\left(1 - \frac{L_a}{W}\right)} - \frac{M_a}{LW}}{\left(1 - \mu_T \frac{h}{L}\right)}
$$

RWD
- tractive efficiency ↑ with ↑ μ_T
- tractive efficiency ↑ with ↑ c.g. height
- tractive efficiency ↑ with ↑ W_r (or l_1)

  - FWD

$$
\frac{F_{T,\max FWD}}{\mu_T W} = \frac{\left(\frac{l_2}{L} + \cancel{\frac{h\mu_r}{L}}\right)\cancel{\left(1 - \frac{L_a}{W}\right)} + \frac{M_a}{LW}}{\left(1 + \mu_T \frac{h}{L}\right)}
$$

FWD
- tractive efficiency ↑ with ↑ μ_T
- tractive efficiency ↓ with ↑ c.g. height
- tractive efficiency ↑ with ↑ W_f (or l_2)

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Tractive Force

- The total tractive force of the system is

$$F_T = F_{Tf} + F_{Tr}$$

- For FWD, F_{Tr}=0
  - Load transfer due to forward acceleration shifts load from the front wheels to the rear wheels
  - This reduces the maximum traction that the front wheels can develop
  - Maximum traction for FWD is before load transfer occurs
- For RWD, F_{Tf}=0
  - Load transfer onto the rear wheels due to forward acceleration increases the maximum traction that the rear wheels can develop
  - The maximum traction lags as first a forward acceleration must be developed for load transfer to occur
- For 4WD (xWD) then both axles provide tractive effort
  - The total traction available, neglecting aerodynamic loads, is available at all times regardless of load transfer
  - However, the total traction available at the front and rear will vary with load transfer effects
  - To maximize total traction the torque split to the front and rear must be controlled and proportional to the normal loads

Components of Resistance

- When determining the resistance to forward motion we need to consider several components
  - Aerodynamic drag
  - Wheel or rolling resistance
  - Resistance due to climbing a grade
  - Acceleration resistance
    - linear and rotational
- From the sum of these resistances we can determine the required force, and power, required

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Aerodynamic Forces

- Aerodynamic forces we are generally interested in are
  - Drag
  - Lift or downforce
  - Pitching Moment
- When incorporating aerodynamic forces into your calculations you must be very careful as to where your data is referenced to
  - All forces and moments at the c.g.?
  - All forces acting at the zero moment point
  - All forces and moments reference to ground plane and pitch moment as %Front (common for racecars with minimal c.g. movement)
  - Other…

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Aerodynamic Forces

- For our resistance calculations we, in general, will need only our drag values.
- Recall that we generally discuss the aerodynamic drag as:

$$F_d = C_D \frac{1}{2} \rho \mathrm{V}^2 A_f$$

where C_D is the drag coefficient, A_f is the frontal area and V is the air velocity

- The power to overcome the aerodynamic drag (or any resistive force) is:

$$P_D = F_D V = C_D \frac{1}{2} \rho \mathrm{V}^2 V A_f = C_D \frac{1}{2} \rho \mathrm{V}^3 A_f$$

- Aerodynamic forces have a huge impact on vehicle performance at higher speeds
  - Lift and drag f(V^2)
  - Power to overcome drag f(V^3)!

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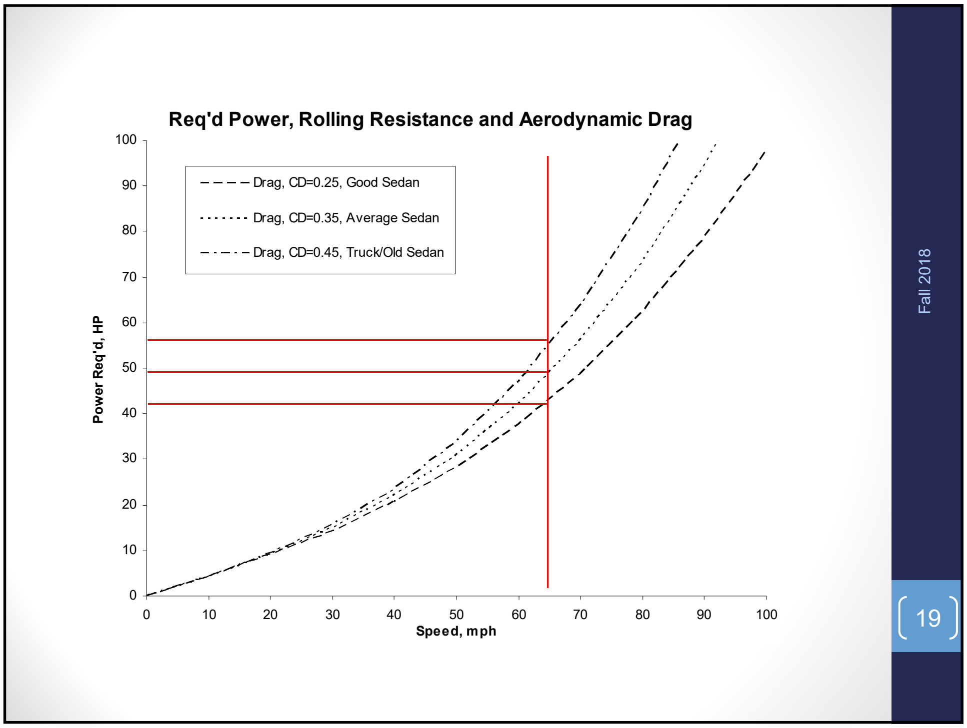
Req'd Power, Rolling Resistance and Aerodynamic Drag
Drag, CD=0.25, Good Sedan
Drag, CD=0.35, Average Sedan
Drag, CD=0.45, Truck/Old Sedan
Power Req'd, HP
Speed, mph
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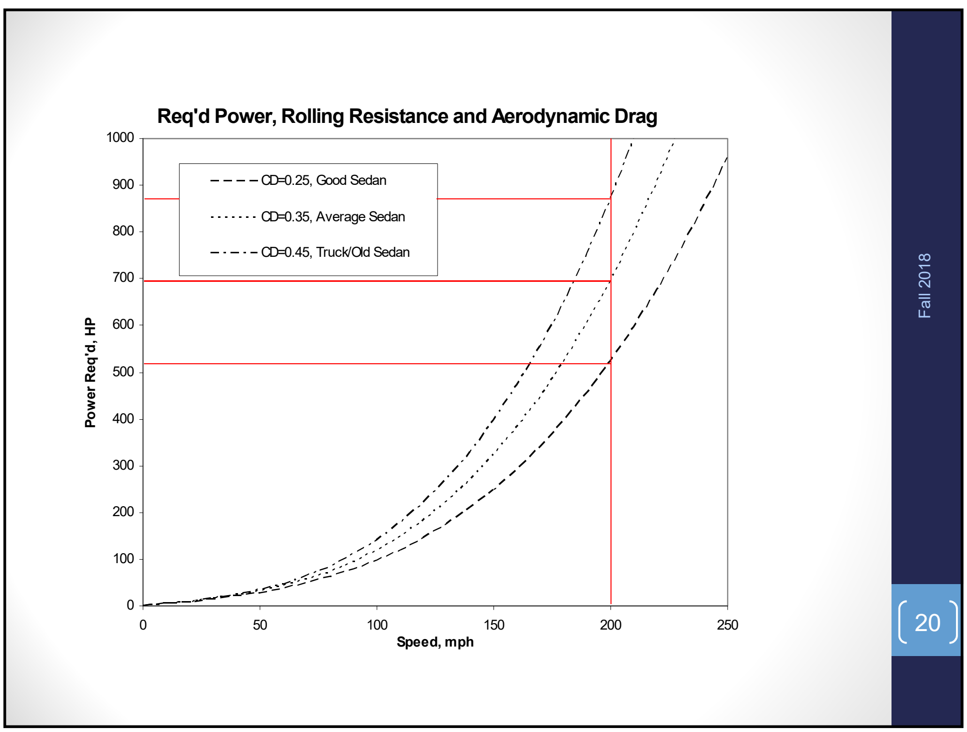
Req'd Power, Rolling Resistance and Aerodynamic Drag
CD=0.25, Good Sedan
CD=0.35, Average Sedan
CD=0.45, Truck/Old Sedan
Power Req'd, HP
Speed, mph
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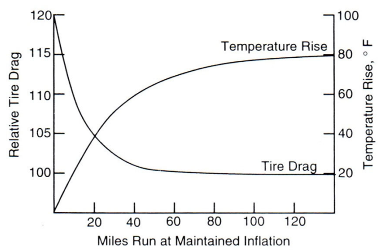
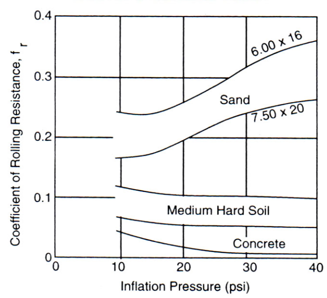
Rolling Resistance





- We discussed rolling or wheel resistance in detail earlier
- We can often estimate rolling or wheel resistance with a simpler mdel with the tires rolling on the ground are treated as a friction effect

$$R_{rf} = \mu_f W_f$$
$$R_{rr} = \mu_r W_r$$

- Rolling resistance is a function of
  - Surface type, i.e. asphalt, dirt, sand
  - Tire construction
  - Tire inflation
  - Slip angle/lateral loads due to construction or alignment issues
- For our purposes in this topic we will use this simpler model

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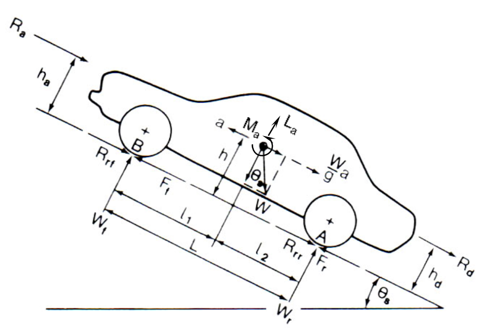
Gradients



- When traveling on a slope or gradient the weight of the vehicle has a components perpendicular and parallel to the road
- The component resisting forward motion is

$$F_g = mg \sin\theta$$

- The grade or incline is specified as
  - Rise over run or slope of the ground

$$grade = \frac{rise}{run}$$

for example,

$$10\%\ grade = \frac{10ft\ rise}{100ft\ run} * 100$$

  - Positive for a increase in height, i.e. uphill
  - Negative for a decrease in height, i.e. downhill
- Grade is often specified by rules and regulations
  - Highways are limited to 6 or 7% (should be confirmed – Grapevine grade?)
  - Local roads can be higher but often limited to 10%
  - SLO driveways are limited to 20%

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Acceleration Resistance

- If we sum the previous resistive loads (aerodynamic, rolling, gradient) then we have the total (assuming trailering loads or whatever are included) load that must be overcome to maintain a given speed
  - This allows us to determine the required steady state power requirements
- If we add in an acceleration term then we can also determine the transient or acceleration load (power) requirments

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Acceleration Resistance, Translational

- The acceleration resistance, $F_{a,trans}$, consists of a linear and rotational component, $F_{a,rot}$
- Translational

$$F_{a,trans} = m * a$$

  - Where the acceleration is the linear acceleration in the positivie x-axis direction or

$$F_{a,trans} = m * \ddot{x}$$

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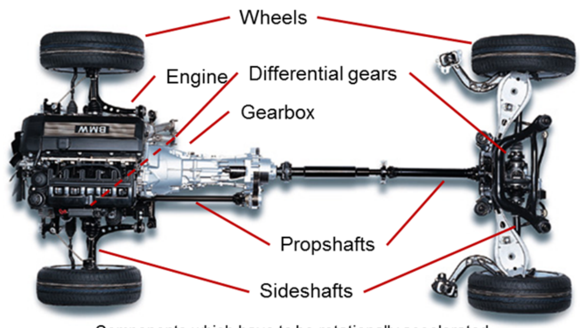
Acceleration Resistance, Rotational

- The rotational component, $F_{a,rot}$, arises from having to accelerate all of the rotating parts of the vehicle
  - Tires/wheels
  - Shafts
  - Engine
  - Etc.



Components which have to be rotationally accelerated

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Acceleration Resistance, Rotational (2)

- The rotational component, $F_{a,rot}$, arises from having to accelerate all of the rotating parts of the vehicle
  - Tires/wheels
  - Shafts
  - Engine
  - Etc.

$$T_{a,rot} = \Theta_{red} * \ddot{\varphi}$$

And

$$T_{a,rot} = F_{a,rot} * r_{dyn}$$

$$F_{a,rot} = \frac{\Theta_{red} * \ddot{\varphi}}{r_{dyn}}$$

Where

$T_{a,rot}$ (Nm or ftlb) is the resistance torque
Θ_{red} (kg m2 or lbf s2 ft) is the moment of inertia reference to the wheels
$\ddot{\varphi}$ (1/s2) is the angular acceleration
r_{dyn} (m or ft) is the dynamic tire radius

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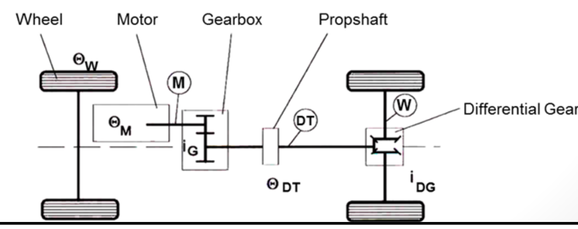
Acceleration Resistance, Rotational (3)

- We generally have three systems with different rotating speeds
  - M: engine, input shaft to gearbox etc. at engine speed (n_M)
  - DT: drivetrain consisting of output shaft of gearbox, input to differential etc. at output speed of transmission (n_T)
  - W: output of differential, drivehafts, wheels etc. at wheel speed (n_W)

  - Since we have differing rotational speeds we must correct to a common (wheel) with

$$\xi_D = \frac{n_T}{n_W}$$

$$\xi_T = \frac{n_M}{n_T}$$



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Acceleration Resistance, Rotational (4)

- Total acceleration resistance

$$F_a = F_{a,trans} + F_{a,rot}$$

$$= m * \ddot{x} + \frac{\Theta_{red} * \ddot{x}}{{r_{dyn}}^2}$$

$$= m * \ddot{x} * \gamma$$

Where γ is the effective mass term by which the mass of the vehicle appears to be increased due to accelerating the rotating components

$$\gamma = 1 + \frac{I_{eff}}{m * {r_{dyn}}^2}$$

$$\gamma = 1 + \frac{I_W}{m * {r_{dyn}}^2} + \frac{I_{DT} * {\xi_D}^2}{m * {r_{dyn}}^2} + \frac{I_M * {\xi_D}^2 * {\xi_T}^2}{m * {r_{dyn}}^2}$$

- γ depends on the gear ratio and therefore depends on the engaged gear. In general, the mass is the unladen vehicle and any additonal mass (passengers, cargo, etc.) will need to be included

$$F_{a,trans} = (m_{unladen} * \gamma + m_{add}) * \ddot{x}$$

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Acceleration Resistance, Rotational (5)

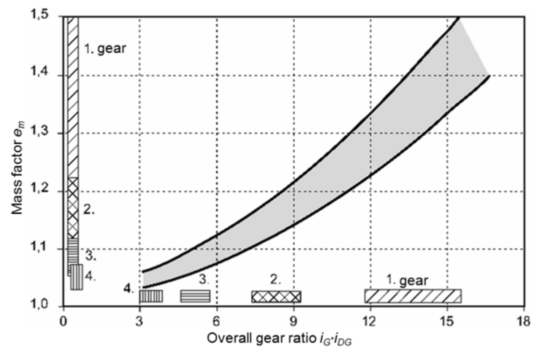
- For

$$\gamma = 1 + \frac{I_W}{m * r_{dyn}{}^2} + \frac{I_{DT} * \xi_D{}^2}{m * r_{dyn}{}^2} + \frac{I_M * \xi_D{}^2 * \xi_T{}^2}{m * r_{dyn}{}^2}$$

and

$$F_{a,trans} = (m_{unladen} * \gamma + m_{add}) * \ddot{x}$$

Typically,



Mass factor e_m vs. total gear ratio

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